

Task 03 – Detailed Summary of Object Tracking Survey Papers

1. Object Tracking Using Computer Vision: A Review

Introduction

This paper provides a comprehensive overview of object tracking in computer vision. The authors analyze the development of tracking systems from traditional image processing approaches to modern deep learning-based solutions. The review focuses not only on algorithms but also on hardware configurations, datasets, and practical applications. It presents object tracking as a complete system rather than only an algorithmic problem.

Object tracking is defined as the process of locating one or multiple objects across video frames while maintaining their identities over time. The main goal is to estimate object position consistently, even under challenging conditions such as occlusion, lighting variation, and background clutter.

Methodology

The paper organizes object tracking research into structured categories. It explains the general tracking pipeline, which typically includes:

1. Image acquisition
2. Pre-processing
3. Object detection
4. Feature extraction
5. Data association
6. Motion prediction
7. Output tracking results

The review divides tracking approaches into traditional methods and learning-based methods.

Traditional Methods

Traditional tracking methods rely on hand-crafted features and mathematical models. These include:

- Background subtraction
- Optical flow
- Feature-based tracking (e.g., corners, edges, color histograms)
- Kalman filters for linear motion prediction
- Particle filters for non-linear motion estimation

These methods are computationally efficient but often struggle in complex environments.

Machine Learning and Deep Learning Methods

Modern tracking systems rely heavily on deep learning. CNN-based feature extractors improve robustness against illumination changes and appearance variations. Deep trackers learn strong object representations from large datasets and perform better in challenging conditions.

The review highlights how deep learning shifted tracking from manual feature engineering to automatic feature learning.

Tools and Technology

An important contribution of the paper is the classification of tracking systems based on hardware configuration:

Monocular Systems

Use a single RGB camera. These systems are affordable and widely used but lack depth information, making distance estimation difficult.

Stereo and Depth-Based Systems

Use stereo cameras or RGB-D sensors. Depth information improves robustness, especially in robotics and navigation tasks.

Hybrid Systems

Combine cameras with other sensors such as LiDAR, radar, IMU, or GPS. Sensor fusion increases accuracy and reliability, particularly in autonomous vehicles.

The review emphasizes that sensor choice strongly affects tracking performance and system design.

On the software side, technologies include:

- OpenCV implementations
- GPU acceleration
- Deep neural network frameworks

Advances in GPU computing significantly improved real-time tracking performance.

Datasets and Evaluation

The paper explains that progress in tracking has been driven by publicly available datasets. These datasets contain annotated video sequences with bounding box ground truth.

Datasets include challenges such as:

- Occlusion
- Illumination changes
- Scale variation
- Motion blur
- Fast object motion

Evaluation metrics commonly used include:

- Precision
- Recall
- Tracking accuracy
- Robustness
- Speed (frames per second)

Benchmarking ensures fair comparison between tracking methods.

Applications

The review discusses multiple real-world applications:

- Autonomous driving (vehicle and pedestrian tracking)
- Robotics (navigation and obstacle avoidance)
- Video surveillance
- Human activity recognition
- Medical imaging
- Sports analytics

Tracking plays a critical role in systems that require continuous object localization over time.

Future Scope

Despite advancements, several challenges remain:

- Robust tracking under heavy occlusion
- Handling object deformation and shape changes
- Tracking in low-light conditions
- Improving real-time performance without sacrificing accuracy
- Better integration of multi-sensor systems

Future research is expected to focus on improving robustness, scalability, and adaptability across different environments.

2. A Deep Dive into Generic Object Tracking: A Survey

Introduction

This paper focuses specifically on Generic Object Tracking (GOT), also known as Single Object Tracking (SOT). In this problem setting, the target object is given in the first frame, and the system must track it in all subsequent frames.

Unlike the first review, which provides a broad system-level perspective, this paper focuses deeply on algorithmic design and modern neural network architectures.

Methodology

The survey categorizes modern trackers into three major families:

1. Discriminative-Based Trackers

These methods treat tracking as a classification problem. The tracker learns to distinguish the target from the background.

Correlation filter-based trackers fall into this category. They are computationally efficient and suitable for real-time applications.

Strengths:

- Fast
- Efficient
- Good for moderate motion

Weaknesses:

- Sensitive to appearance changes
 - Limited long-term robustness
-

2. Siamese-Based Trackers

These trackers use a Siamese neural network structure. Two identical branches process:

- The template (target from first frame)
- Candidate regions from new frames

The network computes similarity between template and search region.

Strengths:

- Strong feature representation
- High speed
- Widely adopted

Weaknesses:

- Limited adaptability to drastic appearance changes
 - Fixed template limitations in long-term tracking
-

3. Transformer-Based Trackers

Transformer-based trackers use attention mechanisms to model global spatial and temporal relationships. Attention allows the model to focus on important regions.

These trackers capture long-range dependencies better than CNN-only models.

Strengths:

- Strong global context modeling
- Better robustness in complex scenes

Weaknesses:

- High computational cost
 - Requires large datasets for training
-

Tools and Technology

Modern trackers rely on:

- Deep CNN backbones (ResNet, EfficientNet, etc.)
- Attention mechanisms
- Transformer architectures
- GPU acceleration

The survey highlights how tracking has evolved from hand-crafted features to large pre-trained neural networks.

Datasets and Benchmarks

The survey emphasizes standardized evaluation using popular tracking benchmarks. These datasets include:

- Long sequences

- Severe occlusion
- Background clutter
- Fast motion
- Scale and rotation variation

Evaluation metrics include:

- Success rate
- Precision
- Accuracy
- Speed

The paper also discusses the trade-off between accuracy and efficiency. Some models achieve high accuracy but are not suitable for real-time applications.

Applications

Generic object tracking is widely used in:

- Surveillance systems
- Autonomous vehicles
- Robotics
- Augmented reality
- Medical monitoring
- Human-computer interaction

Reliable tracking improves performance in automated decision systems.

Future Scope

The survey identifies open research challenges:

- Long-term tracking with re-detection capability
- Handling distractor objects
- Improving robustness to extreme occlusion
- Reducing computational complexity of transformer-based models
- Improving generalization across different datasets

Future research is expected to focus on efficient transformer architectures and improved adaptability.

Overall Discussion

The first paper provides a broad overview of object tracking systems, including sensors, hardware, algorithms, datasets, and applications. It gives a system-level understanding of tracking technology.

The second paper focuses deeply on modern tracking architectures and provides a detailed classification of discriminative, Siamese, and transformer-based trackers. It analyzes the evolution of tracking methods and compares performance trade-offs.

Together, these two papers provide both theoretical and practical understanding of object tracking. They show how the field evolved from classical image processing techniques to deep learning-based methods and how future research will likely focus on robustness, efficiency, and long-term tracking performance.